

Amar Telidji University Laghouat

Department of Computer Science

Level: 4th Year Engineering

Course: Advanced Network

Lab 01: Wireshark Lab (UDP)

Supervised by:

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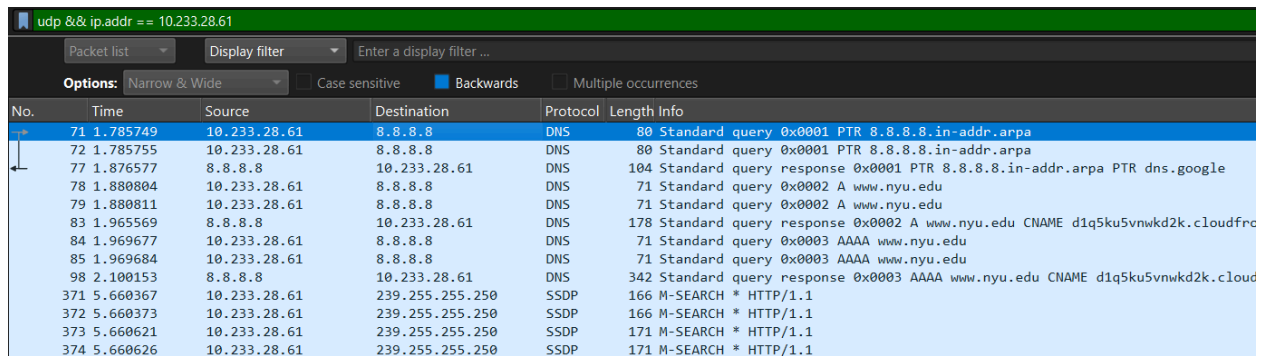
Prepared by:

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Academic Year: 2025 / 2026

Introduction:

The purpose of this lab is to analyze the User Datagram Protocol (UDP) using Wireshark and understand its structure, function, and role in network communication. By capturing and examining real UDP packets generated through DNS queries, this experiment demonstrates how UDP provides simple, connectionless data transfer with minimal overhead. Through inspection of packet fields, header lengths, and request–reply behavior, we gain practical insight into how UDP supports various Internet applications.

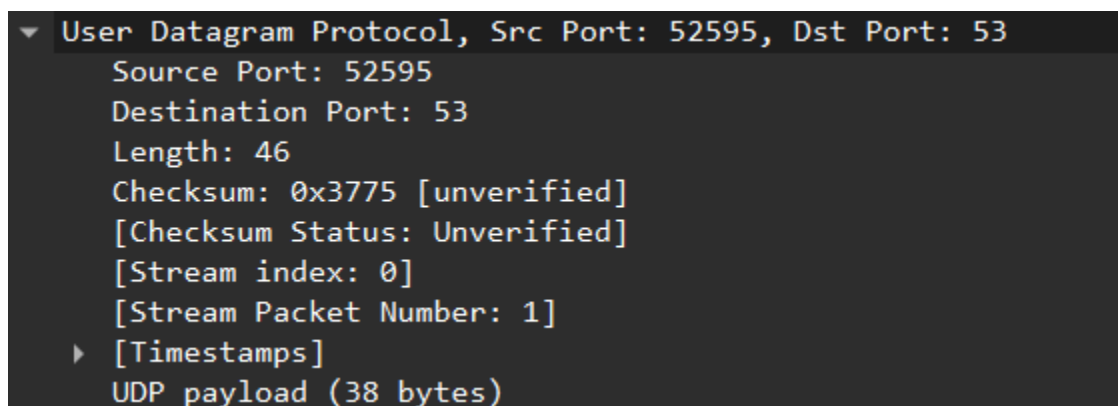


No.	Time	Source	Destination	Protocol	Length	Info
71	1.785749	10.233.28.61	8.8.8.8	DNS	80	Standard query 0x0001 PTR 8.8.8.8.in-addr.arpa
72	1.785755	10.233.28.61	8.8.8.8	DNS	80	Standard query 0x0001 PTR 8.8.8.8.in-addr.arpa
77	1.876577	8.8.8.8	10.233.28.61	DNS	104	Standard query response 0x0001 PTR 8.8.8.8.in-addr.arpa PTR dns.google
78	1.880804	10.233.28.61	8.8.8.8	DNS	71	Standard query 0x0002 A www.nyu.edu
79	1.880811	10.233.28.61	8.8.8.8	DNS	71	Standard query 0x0002 A www.nyu.edu
83	1.965569	8.8.8.8	10.233.28.61	DNS	178	Standard query response 0x0002 A www.nyu.edu CNAME d1q5ku5vwnkd2k.cloudfr
84	1.969677	10.233.28.61	8.8.8.8	DNS	71	Standard query 0x0003 AAAA www.nyu.edu
85	1.969684	10.233.28.61	8.8.8.8	DNS	71	Standard query 0x0003 AAAA www.nyu.edu
98	2.100153	8.8.8.8	10.233.28.61	DNS	342	Standard query response 0x0003 AAAA www.nyu.edu CNAME d1q5ku5vwnkd2k.cloud
371	5.660367	10.233.28.61	239.255.255.250	SSDP	166	M-SEARCH * HTTP/1.1
372	5.660373	10.233.28.61	239.255.255.250	SSDP	166	M-SEARCH * HTTP/1.1
373	5.660621	10.233.28.61	239.255.255.250	SSDP	171	M-SEARCH * HTTP/1.1
374	5.660626	10.233.28.61	239.255.255.250	SSDP	171	M-SEARCH * HTTP/1.1

1. Packet number: 71

Application protocol (payload): DNS (Domain Name System)

The UDP header consists of 4 fields (**Source Port, Destination Port, Length, Checksum**).



2. Each of the UDP fields (**Source Port, Destination Port, Length, Checksum**) is **2 bytes** long. The total header length is **8 bytes**.

3. The UDP Length field (**46 bytes**) represents **the total number of bytes of the UDP header (8 bytes) and payload (38 bytes) combined**.
4. UDP Length field = (2 bytes) 16 bits → Maximum datagram size = $2^{16} - 1 =$ **65,535 bytes**
Subtract 8-byte header → **Maximum UDP payload = 65,527 bytes**
5. UDP port field = 16 bits
Maximum port number = **65,535**

```
▼ Internet Protocol Version 4, Src: 10.233.28.61, Dst: 8.8.8.8
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 66
    Identification: 0x8210 (33296)
  ▶ 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.233.28.61
    Destination Address: 8.8.8.8
    [Stream index: 22]
```

6. The IP Protocol field for UDP is **17**.

71	1.785749	10.233.28.61	8.8.8.8	DNS
72	1.785755	10.233.28.61	8.8.8.8	DNS
77	1.876577	8.8.8.8	10.233.28.61	DNS

Source Port: 52595 Destination Port: 53	Source Port: 53 Destination Port: 52595
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7. The request (Packet 71) is from 10.233.28.61 to 8.8.8.8 (Source Port: 52595, Dest Port: 53).
The reply (Packet 77) is from 8.8.8.8 to 10.233.28.61 (Source Port: 53, Dest Port: 52595).
The ports are swapped, confirming a request–reply pair.

Conclusion:

From this lab, it was observed that UDP operates as a lightweight, connectionless transport protocol with an 8-byte header containing only four essential fields. The captured DNS packets verified that UDP encapsulates application data without establishing connections, offering faster transmission at the cost of reliability. The experiment also confirmed theoretical values such as the maximum payload size, port number range, and protocol number (17). Overall, this lab reinforced the fundamental characteristics and simplicity of UDP in real network communication.